University of Pisa



Business and Project Management

EducationSphere

Authors:
Mattia Correnti
Francesco Ladu

Professors:
Prof. Andrea Bonaccorsi
Prof. Filippo Chiarello

Contents

1 Problem–Solution and Product–Market Fit **4**
- 1.1 Defining the Problem . . . 4
  - 1.1.1 Clearly Articulate the Specific Problem and Identify Who Experiences It . . . 4
  - 1.1.2 Why Current Solutions Fall Short . . . 4
- 1.2 AI-Enabled Solution . . . 5
  - 1.2.1 Personalization . . . 5
  - 1.2.2 Superior Performance . . . 5
  - 1.2.3 Scalability . . . 6
  - 1.2.4 Main Features of EducationSphere . . . 6
- 1.3 Achieving Product–Market Fit . . . 7
  - 1.3.1 Identifying the Target Customer Segment . . . 7
  - 1.3.2 Demonstrating Why the Solution Meets Needs Better Than Alternatives . . . 7

2 Competitors and Positioning **9**
- 2.1 Competitive Analysis . . . 9
- 2.2 Competitive Landscape . . . 9
  - 2.2.1 Direct Competitor: AI-native startups . . . 9
  - 2.2.2 Incumbents . . . 10
  - 2.2.3 Indirect Competitors . . . 10
  - 2.2.4 Substitute Solution Without AI . . . 10
- 2.3 Strategic Positioning and Validation of Competitiveness . . . 11
- 2.4 Patent Landscape . . . 11

3 Business Model Canvas **12**
- 3.1 Business Model Canvas . . . 12
  - 3.1.1 Customer Segments . . . 12
  - 3.1.2 Value Propositions . . . 12
  - 3.1.3 Channels . . . 13
  - 3.1.4 Customer Relationships . . . 14
  - 3.1.5 Revenue Streams . . . 14
  - 3.1.6 Key Resources . . . 14
  - 3.1.7 Key Activities . . . 15
  - 3.1.8 Key Partners . . . 15
  - 3.1.9 Cost Structure . . . 16
- 3.2 Feasibility, Assumptions and Scalability . . . 17

4 Addressable Market Analysis **19**
4.1 Market Size Components . . . 19
4.2 Estimation Methodology . . . 19
4.3 Investment Decision Support . . . 20

5 Personas and Validation Strategy **22**
5.1 User Personas Definition . . . 22
5.1.1 Persona 1: The Problem-Solving Learner . . . 22
5.1.2 Persona 2: The Exam-Ready Student . . . 22
5.1.3 Persona 3: The Accessibility-Sensitive Learner . . . 23
5.2 Survey Sample Overview . . . 23
5.2.1 Level of University Education . . . 24
5.2.2 Field of Study . . . 24
5.2.3 Current Use of AI Tools . . . 24
5.2.4 Satisfaction with Current AI Chatbots . . . 25
5.2.5 Limitations of Current AI Chatbots . . . 25
5.2.6 Main Difficulty During Exam Preparation . . . 26
5.2.7 Feature Validation . . . 26
5.2.8 Expected Price Compared to Generic AI Chatbots . . . 27
5.2.9 Preferred Payment Model . . . 28
5.3 Validation of the Initial Personas . . . 28
5.3.1 Limitations of the Survey . . . 29
5.4 Validation of the Third Persona: The Accessibility-Sensitive Learner . . . 29
5.4.1 Updates needed for our application . . . 31
5.5 Additional Validation Methods . . . 31
5.5.1 Semi-Structured Interviews with Students . . . 31
5.5.2 Clickable Prototype Test . . . 32
5.5.3 MVP Test with a Small Group of Students . . . 32
5.5.4 Institutional Validation with Universities and CRUI . . . 32
5.6 Learning and Decision-Making . . . 33
5.6.1 Success Signals . . . 33
5.6.2 Failure Signals . . . 33
5.6.3 Evidence-Based Product Decisions . . . 34

6 Failure Analysis and Risk Mitigation **35**
6.1 Negative Backcasting Method . . . 35
6.2 Common Failure Causes . . . 35
6.2.1 Strategic Misalignment . . . 35
6.2.2 Market Adoption Issues . . . 36
6.2.3 Technological and Reliability Risks . . . 36
6.2.4 Ethical Barriers . . . 37
6.2.5 Business Model and Cost Risks . . . 38
6.3 Conclusion . . . 38

A Patents **40**
A.1 Patent and Intellectual Property Landscape . . . 40
A.1.1 Patent Mapping by Platform Component . . . 40
A.1.2 Licensing Strategy and Cost Implications . . . 41
A.1.3 Patent Density as a Signal of Market Maturity and Competition . . . 41

A.2 Queries 42

B Full Interview **43**
B.1 Validation of the Third Persona: Full Interview 43

Chapter 1

Problem–Solution and Product–Market Fit

1.1 Defining the Problem

1.1.1 Clearly Articulate the Specific Problem and Identify Who Experiences It

The primary target audience for our AI-powered study assistant is **university students**. In today's fast-paced academic environment, effective time management has become a critical challenge. Students are continually burdened by multiple complex subjects, tight deadlines, and mounting academic pressure. Many students still rely on traditional time management methods, such as manual planners, fixed schedules, or basic digital calendars. These static systems l**ack the personalization and adaptability needed to handle the highly dynamic nature of a university student's life.** Consequently, when study sessions are missed, or new assignments unexpectedly arise, these rigid planners cannot automatically recalibrate, frequently leading to overburdening, procrastination, and academic burnout [1].

Furthermore, when students struggle with specific course materials, accessing immediate and approachable academic support presents a major bottleneck. Limited instructor office hours and the potential intimidation of approaching teaching assistants can significantly delay the learning process. Recent evaluations of AI-powered learning assistants in higher education underscore a **profound demand for accessible, on-demand support systems**. Research indicates that students highly value AI educational assistants for their convenience and comfort; in fact, nearly half of surveyed students reported greater ease and less anxiety when consulting an AI tool than when seeking help from human instructors or teaching assistants [2].

1.1.2 Why Current Solutions Fall Short

While the market is currently saturated with educational applications and digital planners, the vast majority function as passive, one-size-fits-all utilities. Traditional time-management tools require continuous manual updates, and standard learning management systems (LMS) typically present uniform study materials to all users, completely disregarding individual proficiency levels, varying schedules, and learning preferences.

Current solutions fundamentally fall short because they lack **intelligent, dynamic responsiveness**. Recent empirical research that used structural equation modeling to analyze AI-powered adaptive learning platforms demonstrates that learner engagement, the primary driver of improved academic outcomes, is heavily contingent on user satisfaction with dynamic platform features. Specifically, without sophisticated mechanisms such as dynamic content adaptation and real-time, personalized feedback, students quickly lose motivation and struggle to achieve meaningful skill progression [3]. Because current digital tools fail to integrate these adaptive interventions, they cannot adjust study plans in real-time or tailor assessments (such as quizzes and summaries) based on ongoing student performance.

1.2 AI-Enabled Solution

The proposed solution is an AI-powered study assistant designed to help university students organize and improve their exam preparation. The role of artificial intelligence is central because it allows the platform to go beyond traditional study planners and static productivity tools.

1.2.1 Personalization

Artificial intelligence enables a high level of personalization by adapting the learning experience to the individual needs, goals, and constraints of each student.

Generative AI and Retrieval-Augmented Generation (RAG) are particularly relevant for this function; they can support personalized learning by creating tailored learning paths, providing real-time assistance, and generating educational content adapted to the learner's level and interests [4]. This is important for our business idea because university students do not all study in the same way: some need more revision time, others need more practice questions, while others need simplified explanations or summaries of complex materials.

In our solution, RAG can also improve the relevance of AI-generated outputs by **connecting the model's answers to the student's own course materials**. This means that summaries, explanations, and quizzes are not generic, but **are based on the actual lecture notes, slides, PDFs, and documents uploaded by the student**. As a result, the platform can offer a more accurate and course-specific study experience.

1.2.2 Superior Performance

AI provides superior performance by transforming students' academic materials into structured and actionable study resources more efficiently than manual study methods. This is supported by the study of Nor Afian et al., which presents an AI-powered tutoring system for automatic notes summarization and quiz generation. The system allows students to upload academic materials such as PDFs, text files, and PowerPoint presentations, which are then processed by natural language processing technologies to generate summaries and quizzes [5]. This supports the design of our AI-powered study assistant. The superior performance of the solution does come from the combination of **speed, structure, and learning usefulness**. The AI can process large quantities of study material faster than a student could do manually, identify the most relevant information, and convert it into active learning tools. Summaries help students review concepts more efficiently, while

quizzes promote self-assessment and active recall. In this way, the platform improves the quality of study sessions by helping students move from passive reading to active preparation.

Moreover, AI allows the platform to provide immediate feedback. After completing quizzes, students can receive explanations of wrong answers, identify weak areas, and update their study plan accordingly.

1.2.3 Scalability

AI also makes the solution scalable by providing personalized academic support to many students simultaneously. Traditional human tutoring is valuable, but it **is limited by availability, cost, location, and time**. A tutor can only support a limited number of students, usually during fixed hours. An AI-powered study assistant, instead, can be accessed on demand and can support many users simultaneously, making personalized study assistance more widely available.

"New America" highlights that AI-based tutoring systems have the potential to overcome barriers related to **availability, cost, and location** in higher education. Also, once these systems become cost-effective, they could provide affordable, on-demand tutoring opportunities to large groups of students, including students who may not be able to access traditional tutoring because of price, work, caregiving responsibilities, or limited schedules.

For EducationSphere, scalability is a key business advantage. The platform can serve students from different universities, degree programs, and subjects because it is not limited to a fixed catalogue of pre-written courses. This allows the business to expand across academic fields without manually creating separate learning content for every subject. However, scalability must be managed responsibly. AI tutoring systems raise privacy and data-governance concerns because they may process large amounts of student data [6]. For this reason, the proposed solution should include **clear data protection policies, transparent use of uploaded materials, and safeguards** to prevent misuse of personal learning data. In this way, scalability is not only a technical feature, but also a strategic requirement: the platform must be able to grow while maintaining **trust, reliability, and responsible data management**.

1.2.4 Main Features of EducationSphere

EducationSphere addresses students' main study difficulties through five core AI-enabled features:

1. **Course-Grounded Tutor**: AI tutor based on approved course materials, designed to provide course-specific answers, summaries, and quizzes while reducing hallucinations.

2. **Exam Twin AI**: Exam simulation tool that generates realistic written and oral mock exams aligned with the course structure, difficulty, and evaluation logic.

3. **Knowledge Gap Map**: Diagnostic tool that identifies weak topics, concepts, prerequisites, and reasoning steps after quizzes or mock exams.

4. **Adaptive Study Strategy Engine**: Personalized study planner that updates the learning path according to exam date, available time, quiz results, and detected gaps.

5. **Accessible Learning Mode**: Accessibility-oriented mode that adapts explanations, study blocks, summaries, exercises, and formats for students with ADHD, dyslexia, dyscalculia, or other learning needs.

1.3 Achieving Product–Market Fit

1.3.1 Identifying the Target Customer Segment

The primary target customer segment for the proposed solution is university students who need personalized support during their study process.

Recent research confirms that generative AI is already widely used by college students. Guo et al. analyze survey data from **72,615 undergraduate students across 25 universities and colleges** in China and find that more than sixty percent of students used generative AI technologies during the 2023–2024 academic year. This supports the choice of university students as the main customer segment, because the market is already familiar with AI-based learning support, and the educational barrier to adoption is relatively low. However, the same study also shows that the effect of generative AI on student engagement depends on the learning environment. While tools such as ChatGPT and Deepseek can be associated with higher cognitive and emotional engagement in high-challenge and high-support contexts, they may also reduce active learning and motivation in other situations [7]. This suggests that students need AI tools designed to support genuine learning, **not tools that simply provide answers**. The adoption potential of this segment is further supported by Sergeeva et al., who investigate higher education students' adoption of generative AI technologies using the UTAUT2 model. Their findings show that habit is the strongest predictor of actual generative AI adoption, followed by performance expectancy. Social influence and price value also positively influence students' behavioral intention to use these technologies [8].

1.3.2 Demonstrating Why the Solution Meets Needs Better Than Alternatives

The proposed AI-powered study assistant meets student needs better than traditional study methods because it combines planning, personalization, and content processing in a single adaptive system. Traditional methods, such as handwritten notes, static timetables, printed summaries, and manual flashcards, can be useful but require substantial time and self-discipline. They also depend heavily on the student's ability to estimate priorities, organize materials, and identify weak areas without external support.

Ward et al. [9] analyze the impact of AI tools on student study habits and academic performance. Their study reports that students showed strong support for AI-based features and that AI tools were associated with a reduction in study hours and an increase in GPA(Grade Point Average). Although the authors also warn about risks such as over-reliance on AI and difficulties in integrating AI with traditional teaching methods, their findings suggest that AI can improve study efficiency when used as a complement to

conventional learning strategies.
Compared with traditional methods, our solution offers three main advantages.

1. It reduces the time required to organize study activities. Instead of manually creating a study calendar, students can receive a personalized plan based on exam dates, available time, subject difficulty, and progress.

2. It transforms passive learning materials into active study resources, such as summaries and quizzes. This helps students move beyond simple reading and supports active recall and self-assessment.

3. It provides continuous feedback through mock exams and oral exam simulations, allowing students to understand where they are improving and where they still need revision.

The proposed platform also improves on generic AI tools. Many students already use general-purpose chatbots such as Gemini or ChatGPT, but **these tools are not specifically designed for exam preparation or long-term study management**. A generic chatbot may answer questions or summarize a document, but it does not necessarily create a structured preparation path, track progress over time, reschedule missed study sessions, or connect quiz results to future planning. Our solution is therefore positioned not as a replacement for studying, but as **an integrated study-management assistant.**

This positioning is important because the literature also highlights the limits and risks of AI adoption. Guo et al. [7] show that generative AI can support engagement, but its benefits depend on the learning environment and instructional design. Ward et al. [9] similarly emphasize that AI should complement conventional educational strategies rather than replace them. Therefore, the product-market fit of our solution depends **on offering AI in a controlled, learning-oriented way: the platform should guide students, help them practice, and support reflection, rather than simply generate ready-made answers.**

Chapter 2

Competitors and Positioning

2.1 Competitive Analysis

The competitive analysis of EducationSphere is based on three key assumptions.

1. EducationSphere operates in the AI-powered EdTech market, a growing and increasingly competitive sector. According to Dimension Market Research, the global AI in EdTech market is expected to reach USD 98.1 billion by 2034 [10].

2. EducationSphere does not compete only with AI-native startups. Its competitive environment also includes established EdTech companies,and non-AI substitutes such as tutors, textbooks and official institutional resources [11], [12].

3. The key competitive dimension is not the mere presence of AI, but the ability to provide reliable and pedagogically useful academic support. Competitors must therefore be assessed according to how effectively they help students understand content, solve exercises, prepare for exams, and improve academic performance [13].

2.2 Competitive Landscape

EducationSphere's competitive landscape can be divided into four categories: AI-native startups, incumbents, indirect competitors, and substitute solutions without AI.

2.2.1 Direct Competitor: AI-native startups

AI-native startups represent a strong competitive threat because they are designed around AI-based learning support. Since AI is becoming a standard feature in EdTech, EducationSphere cannot differentiate itself simply by using artificial intelligence. It must instead compete through reliable explanations, course-specific personalization, and continuous support rather than isolated answers [10].

An example of an AI-native competitor is **Knowunity** [14], a learning platform focused on digital study support and peer-based academic content. Its main strengths are its strong focus on students, mobile accessibility, and ability to provide fast study resources in formats that are familiar to younger users. As an AI-enabled learning platform, Knowunity also represents the type of competitor that can quickly integrate generative AI features such as summaries, explanations, and personalized assistance.

However, from EducationSphere's perspective, Knowunity is less centered on each student's own university course materials and exam preparation process. While it can support learning through shared resources and notes given by other students and digital tools, it may offer less control over course-specific grounding, individual study planning, and continuous progress adaptation.

This creates an opportunity for EducationSphere to differentiate itself through course-grounded personalization. Instead of offering general study support, EducationSphere can transform each student's materials into a structured preparation path with adaptive quizzes, feedback loops, and progress tracking.

2.2.2 Incumbents

Incumbents represent a major competitive threat because they combine established brands, large user bases, proprietary educational content, institutional relationships, and the financial resources needed to integrate AI into their existing platforms [11]. A relevant example is **Chegg**, whose traditional model is based on expert-verified solutions. However, the rise of generative AI has challenged this model by making instant and low-cost academic assistance more accessible to students [12]. For EducationSphere, this shows that scale, brand recognition, and content ownership are important competitive advantages, but they are not sufficient to guarantee long-term differentiation. Incumbents may be strong in providing answers and large libraries of educational resources, but they can be less focused on transforming each student's own course materials into a personalized exam preparation process. EducationSphere can therefore differentiate itself by avoiding the positioning of a static answer repository. Instead, the platform should emphasize guided learning and course-specific personalization. This would make the platform more aligned with students' actual preparation process, rather than simply helping them find isolated answers.

2.2.3 Indirect Competitors

Indirect competitors are tools that are not dedicated EdTech platforms but still satisfy student learning needs. The most relevant examples are general-purpose generative AI tools (namely chatbots), search engines, video platforms and online forums. Among them, **ChatGPT** is a particularly relevant indirect competitor because students can use it for many different things: explanations, summaries, coding support, and exam preparation. ChatGPT's main strengths are accessibility, speed, flexibility, and low cost. It has also changed student expectations: academic support is now expected to be immediate. However, ChatGPT is not specifically designed as a structured university study-management platform. Its answers may be too generic or not fully aligned with specific exam requirements.[15] EducationSphere can therefore differentiate itself by transforming the flexibility of generative AI into a controlled, education-specific learning system. The platform should not compete with ChatGPT only on answer generation, because that would be difficult and strategically weak. Instead, it should compete by offering a structured learning workflow based on course material grounding and exam-oriented preparation.

2.2.4 Substitute Solution Without AI

Non-AI substitutes remain relevant because not all students rely on artificial intelligence for academic support. These include textbooks, lecture slides, handwritten notes, and

standard learning management systems [13]. Among them, **human tutoring** is a particularly relevant substitute because it offers personalized explanations and direct interaction. It remains valuable because it can address aspects of learning that are difficult to automate: a human tutor can recognize student anxiety and can adapt the explanation style according to the student's reactions. However, human tutoring also presents important limitations from a business and scalability perspective. It is usually expensive, depends on tutor availability, and is limited to scheduled sessions. "Estha" reports that traditional human tutoring typically ranges from $25 to $150 per hour, while AI tutoring platforms often operate through subscription models ranging from $10 to $50 per month for broader and more continuous access. The same source estimates that a student requiring 10 hours of human tutoring per month may spend approximately $500 to $1,500, creating a high recurring cost for many families [16]. This cost structure makes human tutoring less accessible for students who need frequent support, late-night clarification, or help across multiple subjects.

EducationSphere does not need to replace human tutoring completely. Instead, it can position itself as a more accessible and continuous complement to traditional academic support. This positioning is consistent with the hybrid tutoring logic described by Estha, where AI can support daily practice and instant feedback, while human tutors remain more suitable for complex topics, emotional support, and personalized guidance [16]. For this reason, human tutoring and other non-AI substitutes represent a medium-level competitive threat. They remain strong in terms of trust and human interaction, but EducationSphere can compete through affordability, availability, and scalability.

2.3 Strategic Positioning and Validation of Competitiveness

EducationSphere's positioning is based on five differentiation levers.

1. In terms of customer experience, the platform offers a structured learning workflow rather than isolated answers.
2. In terms of performance, it supports exam preparation through adaptive explanations, quizzes, mock exams, progress tracking, and feedback loops.
3. In terms of cost, it can provide continuous academic support at a lower marginal cost than one-to-one tutoring.
4. In terms of data access, EducationSphere uses uploaded or institutionally approved course materials to generate course-grounded explanations and assessments.
5. In terms of compliance, the platform should emphasize privacy protection, transparent AI use, copyright-aware handling of course materials, and governance controls for universities.

2.4 Patent Landscape

A further analysis of the market for these technologies was conducted through a patent analysis. Patent tags, estimated costs, and the analysis of each query are reported in the dedicated appendix A.

Chapter 3

Business Model Canvas

3.1 Business Model Canvas

3.1.1 Customer Segments

EducationSphere addresses three main customer segments.

1. **University students** preparing for exams. The most attractive early adopters are students who experience academic pressure, poor time management, and exam anxiety. These students need support to organize preparation, practice realistic questions, identify weak topics, and receive explanations based on their own course materials.

2. **Students with learning difficulties or accessibility needs**, such as dyslexia, dyscalculia, ADHD, or other physical conditions (such as blindness) that may increase cognitive load during studying. For these students, EducationSphere can provide short study blocks, micro-goals, simplified summaries, step-by-step explanations, and text-to-speech compatible outputs. This segment connects the platform not only to productivity but also to **inclusion and student support**.

3. The third segment is **institutional**. It includes universities, DSA and disability offices, and tutoring services. They may become paying customers, adoption enablers, or internal promoters.

   This segmentation implies a primarily B2B/B2B2C business logic. Although university students are the final users of EducationSphere, the main paying customers are expected to be universities, departments, tutoring services, DSA and disability offices. In the B2B model, institutions purchase access to the platform through pilot projects or annual licences. In the B2B2C model, institutions act as intermediaries that enable student adoption, increase trust, and reduce the individual willingness-to-pay barrier. Direct student access through a hypothetical freemium model remains relevant, but it should mainly support early adoption, feedback collection, and demand validation rather than represent the core long-term revenue logic.

3.1.2 Value Propositions

The central value proposition of EducationSphere is to **transform exam preparation from a fragmented and passive process into a structured learning experience**.

For students, EducationSphere helps clarify what to study, how to study, which topics require more attention, and how ready they are for the exam.

1. A key element of the value proposition is **course-grounded personalization**. Unlike generic AI tools, EducationSphere generates explanations, summaries, quizzes, and mock exams based on uploaded institutionally approved course materials, such as slides, lecture notes, exercises, textbooks, and previous exam formats. This makes the learning support more aligned with the actual course and reduces the risk of generic or irrelevant answers.

2. EducationSphere also supports active learning. Instead of relying only on passive reading, students can practice with realistic questions, oral-style follow-ups, and mock exams, helping them identify weak areas and decide where to focus their effort.

3. For students with learning difficulties or accessibility needs, EducationSphere provides **accessible and personalized study support** by adapting the format, rhythm, and compl
This can reduce cognitive overload and make exam preparation more manageable.

4. For universities and support offices, the value proposition is **scalability**. EducationSphere can complement tutoring, DSA, disability, and wellbeing services by providing continuous digital support without requiring a proportional increase in staff workload. It can also provide aggregated and privacy-compliant insights about student engagement, recurring learning gaps, and accessibility needs.

3.1.3 Channels

1. **University pilot projects**, which are a credible entry point since they allow the startup to test the platform with real students, collect evidence, and reduce adoption risk. Initial pilots should target DSA and disability offices, tutoring services, or selected departments.

2. Direct student access through a freemium model, supported by a targeted **social media strategy**. EducationSphere can use platforms frequently used by university students, such as Instagram, TikTok, and university-specific digital communities. Marketing content should address core user frustrations, such as exam anxiety and poor time management, while showing short demonstrations of the platform's AI features.

3. Another important channel is **student-to-student referral**. Since university students often rely on peer recommendations when choosing study tools, EducationSphere can encourage existing users to invite classmates, friends, or members of the same course. Referral mechanisms, such as premium feature trials, discounts, or extra study credits, can support organic growth and reduce customer acquisition costs.

4. **University communities and informal student groups** also represent a relevant channel. EducationSphere can reach students through university online communities, Telegram groups, WhatsApp groups, student associations, and course-specific discussion groups. These channels are particularly useful because they are already used by students to exchange notes and exam information.

3.1.4 Customer Relationships

1. **With students**, the relationship should be personalized, continuous, and adaptive. Inside the platform, EducationSphere acts as a study companion that provides feedback, progress tracking, and mock exams during exam preparation.

2. **With students who have accessibility needs**, the relationship must also be supportive and non-judgmental. The platform should avoid overwhelming the user and should provide clear, manageable, and confidence-building study steps.

3. **With universities and support offices**, the relationship should be consultative and evidence-based. EducationSphere must demonstrate measurable value through pilot results, usage analytics, and student satisfaction. Institutional customers also require transparency, data protection, governance controls, and clear boundaries regarding the use of AI.

4. **With university staff**, the relationship should be collaborative rather than substitutive. EducationSphere should not be presented as a replacement for tutors or professors. Instead, it should be positioned as a scalable digital complement that supports routine study preparation and accessibility adaptation, while leaving educational responsibility and governance to the university.

3.1.5 Revenue Streams

The first and most important revenue stream is represented by **institutional pilot contracts and annual university licences**. In the initial phase, EducationSphere should not assume immediate large-scale adoption. A more realistic approach is to enter universities through limited pilot projects with one department, professor, or DSA/Invalidity office. These pilots may be free, low-cost, or paid depending on the institution, but their main purpose is validation rather than short-term revenue. They allow EducationSphere to collect evidence about usage, student satisfaction, and institutional fit.
If pilot results are positive, EducationSphere can convert them into recurring annual licences. Universities may be willing to pay because the platform supports student retention and tutoring capacity. The licence can be structured as a fixed annual fee, a per-student fee, a department-level licence, or a campus-wide agreement. Pricing may also depend on the number of active courses, the number of users, and the accessibility tools included. This stream should become the largest contributor to overall revenues in the medium to long term.

The second revenue stream is a **freemium-to-premium model for individual students**. Many students are price-sensitive and already use free or low-cost alternatives such as ChatGPT, YouTube, shared notes, textbooks, and peer learning. For this reason, EducationSphere should offer a basic free version to attract users and validate demand, while reserving advanced features for the premium version. Premium features may include unlimited document uploads, advanced mock exams, detailed personalized study plans.

3.1.6 Key Resources

1. **Technical development team**: EducationSphere needs software engineers, AI engineers, and product designers to build and maintain the platform. Their role is

to develop the user interface, the AI-based learning features, backend systems, and possible integrations with existing university learning management systems.

2. The second key resource is**educational and accessibility expertise**. To ensure that the platform is pedagogically useful, EducationSphere requires collaboration with instructors and student support professionals. This expertise is especially important for validating both the learning-oriented and accessibility-oriented features for students with dyslexia, dyscalculia, ADHD, invalidities, or other learning difficulties.

3. The third key resource is **institutional relationships**. Since EducationSphere follows a B2B/B2B2C logic, universities and DSA/disability offices are essential resources for pilots, validation, adoption, and long-term licensing.

4. The fourth key resource is **data governance and compliance capability**. EducationSphere processes student data and course materials, which creates privacy, copyright, security, and responsible AI obligations. The company, therefore, needs clear data policies, consent mechanisms, and copyright-aware handling of uploaded materials.

5. The fifth key resource is **scalable software and cloud infrastructure**. The platform must support multiple students, courses, departments, and universities without requiring a proportional increase in staff workload. This includes cloud hosting, model access, storage, cybersecurity, and backup systems.

3.1.7 Key Activities

1. **Platform development**: EducationSphere must continuously develop and improve its features.

2. **Educational validation**: The platform must ensure that its outputs are accurate and appropriate for exam preparation. This may require collaboration with instructors and educational designers.

3. **Pilot management**: Since the initial go-to-market strategy is based on university pilots, EducationSphere must design and evaluate pilot projects. This includes defining success metrics, collecting feedback, and preparing reports for institutional stakeholders.

4. **Data protection and AI governance**: the platform must maintain responsible AI practices, clear data policies, and safeguards against misuse.

5. **Institutional sales and partnership development**: EducationSphere must build relationships with universities, DSA and disability offices, tutoring services, student wellbeing units, departments, and innovation offices.

3.1.8 Key Partners

1. **Universities** are the most important partners because they provide access to students, course contexts, and potential revenue through pilots or licences.

2. **DSA and disability offices** are key partners for accessibility validation and institutional adoption. They can help identify student needs, test accessible learning features, and evaluate whether the platform reduces cognitive overload and improves support scalability.

3. **Technology partners** are necessary for AI infrastructure, cloud hosting and data management.

4. **Accessibility experts and legal consultants** represent another important partner category. Their role is essential for validating accessibility claims and ensuring compliance with privacy and intellectual property requirements.

3.1.9 Cost Structure

- **Software development**: The platform requires continuous investment in backend infrastructure, frontend design, AI workflows, monitoring, and security. Based on industry estimates, building an enterprise-grade EdTech Minimum Viable Product (MVP) requires an initial capital expenditure between $25,000 and $60,000, while expanding to a fully featured platform with advanced AI integrations, analytics, and mobile applications can scale between $70,000 and $150,000+ [17].

- **AI and cloud infrastructure**: EducationSphere may incur costs related to model access, document processing, storage, server capacity, and data processing. Efficient models such as GPT-4o mini cost $0.15 per 1 million input tokens and $0.60 per million output tokens, while mid-to-high tier frontier models range from $2.50 to $15.00+ per million tokens [18]. These costs scale with the number of users, uploaded materials, and generated outputs. Depending on data volume and traffic, cloud infrastructure and database hosting costs are projected to range from $500 to $5,000+ per month.

- **Educational and accessibility validation**: The platform must test whether its features are useful and reliable. This may require collaboration with instructors, students, and disability offices. Professional WCAG compliance audits typically cost $1,500 to $5,000 for standard websites and $7,500 to $21,000+ for feature-rich software platforms, depending on complexity [19].

- **Legal, privacy, and intellectual property compliance**: EducationSphere must manage student-data protection, such as FERPA (Family Educational Rights and Privacy Act) and GDPR (General Data Protection Regulation), copyright issues related to uploaded course materials and freedom-to-operate risks connected to AI tutoring. A comprehensive SaaS compliance framework, including Terms of Service, Privacy Policies, and Data Processing Agreements, typically costs between $1,500 and $5,000 upfront, with ongoing attorney fees averaging $250 to $400 per hour for enterprise contract reviews [20].

- **Marketing and customer acquisition**. To support the direct-to-student freemium channel, EducationSphere should allocate its marketing budget across paid social advertising, short-form content production, creator collaborations, community management, and referral incentives. Paid advertising may include Instagram Reels, Instagram Stories, TikTok in-feed ads, and exam-period campaigns.

  - Instagram advertising costs are generally estimated between $0.40 and $2.00 per click and between $2 and $6 per 1,000 impressions [21] [22].
  - TikTok advertising costs are commonly estimated between $6 and $12 CPM and between $0.50 and $1.50 CPC (Cost per Click)[23].
  - Student creators and micro-influencers may also be used to reach university communities more credibly: nano-influencers may cost around $25 to $150 per sponsored post, while micro-influencers can range from about $250 to $5,000 per post, depending on audience size, engagement, platform, and format [24].

- **Referral incentives**: Referral incentives such as free premium trials, extra document uploads, mock exam credits, or discounted semester plans can support student-to-student growth and reduce reliance on paid advertising. As a benchmark, SaaS referral programs may use revenue-based incentives, sometimes around 20–30% of recurring revenue [25]. Therefore, EducationSphere could initially limit referral costs by offering platform credits or temporary premium access instead of direct monetary rewards.

- **Patent, intellectual property, and licensing costs**: EducationSphere may also incur costs related to patent analysis, freedom-to-operate reviews, and potential licensing agreements for protected technologies connected to its AI-enabled features. Further details are discussed in Appendix A.

3.2 Feasibility, Assumptions and Scalability

The Business Model Canvas suggests that EducationSphere is most feasible when positioned as an institutional AI-based study support platform rather than as a purely consumer application. A B2C model can help attract students, collect feedback, and increase brand awareness, but it also exposes the startup to strong competition, low willingness to pay, and higher customer acquisition pressure. For this reason, **the B2B/B2B2C model based on university pilots and annual institutional licences should remain the main strategic direction**.

The model relies on several critical assumptions.

- Students must perceive exam preparation as a real pain point, especially when they face fragmented materials, poor planning, uncertainty about exam readiness, and a lack of feedback.

- Course-grounded AI support must be perceived as more valuable than generic AI assistants.

- Universities, DSA/disability offices, tutoring services, and student wellbeing units must be willing to test the platform through limited pilots.

- Accessibility must represent a credible differentiator, but this requires expert validation and evidence from real users.

- AI and cloud infrastructure costs must remain compatible with subscription and licence revenues.

EducationSphere should therefore follow a staged growth path.
In the first phase, the startup should **validate the platform through a limited pilot** with one university, department, tutoring service, or DSA/disability office. The objective is to verify repeated student usage, perceived usefulness, accessibility value, and institutional interest.
In the second phase, validated pilots should be **converted into annual institutional licences.**
In the third phase, **the platform can scale across additional departments and universities** by reusing its core AI infrastructure while adapting it to different course materials and exam formats.

Chapter 4

Addressable Market Analysis

4.1 Market Size Components

The addressable market analysis estimates the economic potential of EducationSphere by distinguishing between the Total Addressable Market (TAM), the Serviceable Available Market (SAM), and the Serviceable Obtainable Market (SOM). This distinction separates the theoretical market opportunity from the portion that the company can realistically target and capture in the short term.

The **Total Addressable Market (TAM)** includes European university students and higher education institutions that could potentially benefit from AI-based exam preparation features. According to Eurostat, there were approximately 18.8 million tertiary education students in the European Union in 2023 [26]. From an institutional perspective, Europe has close to 5,000 higher education institutions [27].

The **Serviceable Available Market (SAM)** focuses on the Italian university system, which represents the most realistic first expansion market for EducationSphere. According to MUR/USTAT, Italian universities had 2,026,774 enrolled students in the academic year 2024/2025 [28]. In addition, ANVUR reports that the Italian university system includes 100 universities in 2025 [29].

The **Serviceable Obtainable Market (SOM)** reflects the short-term commercial objective of the project. Since EducationSphere is still at an early stage, the first obtainable market should be based on a limited number of institutional pilots with universities, departments, tutoring services, or DSA/disability offices. These pilots would allow the team to validate demand, measure student engagement, and collect evidence before negotiating larger annual institutional licences.

4.2 Estimation Methodology

The estimation combines a top-down and bottom-up logic. The top-down approach is useful to understand the theoretical market potential, while the bottom-up approach is more realistic for EducationSphere because the business model is based on gradual institutional adoption rather than immediate mass-market subscriptions.

A student-based TAM can be estimated as follows:

$$TAM = Number\ of\ university\ students \times Potential\ annual\ revenue\ per\ student$$

Using the Eurostat figure of 18.8 million tertiary education students in the European Union [26], and assuming a conservative potential annual revenue of 10 per student, the student-based TAM is:

$$TAM = 18,800,000 \times 10 = 188,000,000$$

This estimate does not imply that EducationSphere will sell directly to every student. It only represents the theoretical value of the European student market if AI-based university learning support became widely adopted.

From an institutional perspective, the TAM can be estimated as:

$$TAM = Number\ of\ higher\ education\ institutions \times Average\ annual\ institutional\ licence$$

Considering around 5,000 European higher education institutions [27], and assuming an average annual institutional licence of 20,000, the institutional TAM is:

$$TAM = 5,000 \times 20,000 = 100,000,000$$

For the initial SAM, the Italian university system is used as the first realistic target. Assuming 100 Italian universities [29] and an expected annual licence value of 15,000 per institution, the SAM is:

$$SAM = 100 \times 15,000 = 1,500,000$$

Finally, the SOM reflects a conservative first commercial milestone. Assuming three pilot institutions and an average pilot revenue of 3,000, the SOM is:

$$SOM = 3 \times 3,000 = 9,000$$

This SOM is intentionally conservative. Its purpose is not to represent the long-term potential of the company, but to define a realistic first validation target.

The key assumptions behind this estimation are the following:

1. EducationSphere starts with a B2B or B2B2C model, where institutions pay for access and students use the platform as part of their learning support.
2. The average annual licence varies depending on the number of students, courses, and services included.
3. Pilot projects are priced lower than full licences because their main purpose is validation rather than immediate revenue maximisation.
4. Successful pilots generate the evidence needed to convert early adopters into recurring institutional customers.

4.3 Investment Decision Support

The addressable market analysis supports a cautious but positive investment decision. The potential market is large, with around 18.8 million tertiary education students in the European Union [26] and more than 2 million enrolled university students in Italy [28]. However, the most realistic opportunity is not immediate mass adoption, but gradual institutional adoption through selected pilots.

For this reason, EducationSphere should follow a staged investment logic.

1. In the first stage, the team should build a Minimum Viable Product focused on the most validated features: course-grounded tutoring, quizzes, mock exams, and adaptive feedback.

2. In the second stage, the platform should be tested through one or more institutional pilots.

3. In the third stage, the project should measure usage, satisfaction, perceived learning support, and willingness to pay.

Chapter 5

Personas and Validation Strategy

5.1 User Personas Definition

5.1.1 Persona 1: The Problem-Solving Learner

The first persona is a student who frequently uses AI tools but needs more reliable and structured explanations. This user studies technical subjects and often faces exercises, formulas, problem-solving tasks, and practical applications.

- **Profile:** a Bachelor's or Master's STEM student who already uses AI tools frequently for studying.
- **Goals:** the student wants to understand exercises deeply, receive step-by-step explanations, and practice with problems similar to those in the exam.
- **Frustrations:** generic AI chatbots sometimes provide incorrect answers, do not always explain the reasoning clearly, and are not aligned with the specific materials of the course. This can make the student waste time checking the validity of answers or searching for alternative explanations.
- **Relevant EducationSphere features:**
  - Step-by-step explanation of exercises;
  - AI tutor based on uploaded course materials;
  - Quizzes generated from course materials;
  - Realistic mock exams.

5.1.2 Persona 2: The Exam-Ready Student

The second persona is a student who studies consistently but struggles to understand whether he or she is truly ready for the exam. This user needs realistic practice, feedback, and a clearer view of preparation gaps before the actual exam.

- **Profile:** a university student from either STEM or non-STEM fields who wants to test exam readiness before the actual exam.
- **Goals:** the student wants to practice with realistic exam questions, simulate the exam experience, and identify weak topics before the exam date.

- **Frustrations:** generic chatbots do not reproduce the professor's exam style, do not create realistic exams, and do not provide a clear view of preparation gaps. As a result, the student may study consistently but still feel uncertain about actual exam readiness.
- **Relevant EducationSphere features:**
  - Realistic mock exams based on the course exam style;
  - Knowledge Gap Map showing weak topics;
  - Quizzes generated from course materials;
  - Oral exam simulation with follow-up questions.

5.1.3 Persona 3: The Accessibility-Sensitive Learner

The third persona is a student with learning difficulties or accessibility needs who requires more structured, manageable, and accessible study support. This user may experience difficulties with long materials, dense explanations, a lack of structure, or maintaining focus during exam preparation.

- **Profile:** a university student with learning difficulties or accessibility needs, such as dyslexia, dyscalculia, ADHD, or high cognitive load during studying.
- **Goals:** the student wants to study through shorter and clearer learning steps, receive simplified explanations, follow micro-goals, and reduce cognitive overload during exam preparation.
- **Frustrations**: traditional course materials can be too long, complex, or difficult to navigate. Generic AI tools may provide explanations that are not sufficiently adapted to the student's learning needs, making study sessions confusing or overwhelming. Moreover, some accessibility barriers may depend on physical or sensory impairments, such as blindness, rather than on the student's study method or academic ability. These barriers can make visual materials, graphs, formulas, and non-accessible documents difficult to use.
- **Relevant EducationSphere features**
  - Simplified summaries;
  - Step-by-step explanations;
  - Micro-goals and short study blocks;
  - Visual maps;
  - Text-to-speech compatible outputs;
  - Adaptive study planning.

5.2 Survey Sample Overview

The first two personas and the product assumptions were validated through an anonymous Google Forms survey conducted among university students. The survey collected 100

responses and explored students' current use of AI tools, their main difficulties during exam preparation, the limitations of existing AI chatbots, and their interest in specific EducationSphere features. The survey respondents represent a relevant validation sample because they are university students, which corresponds to the main target user segment of EducationSphere.

5.2.1 Level of University Education

The sample included mainly Master's and Bachelor's students. Specifically, **52% of respondents were Master's students** and **46% were Bachelor's students**, while the remaining respondents belonged to other educational levels.

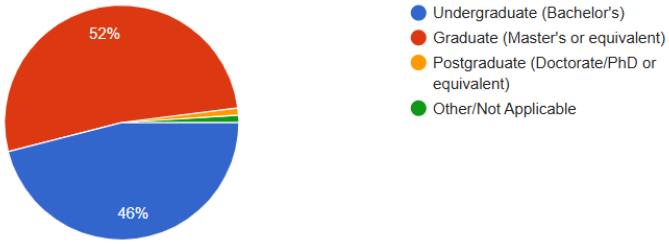


Figure 5.1: Respondents by level of university education.

5.2.2 Field of Study

The sample was strongly represented by **STEM students, who accounted for 58% of respondents**. Other fields included Humanities, Education and Psychology, Medical and Health-related studies, Economics and Management, and Law, Political and Social Sciences.

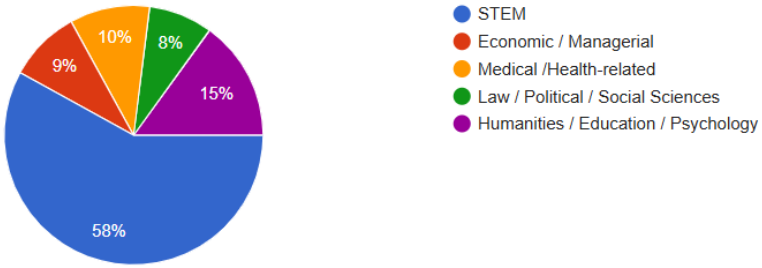


Figure 5.2: Respondents by field of study.

5.2.3 Current Use of AI Tools

A key finding is that **79% of respondents use AI tools frequently, either daily or on most days**, for exam preparation. This shows that the target market is already familiar with AI-based study support.

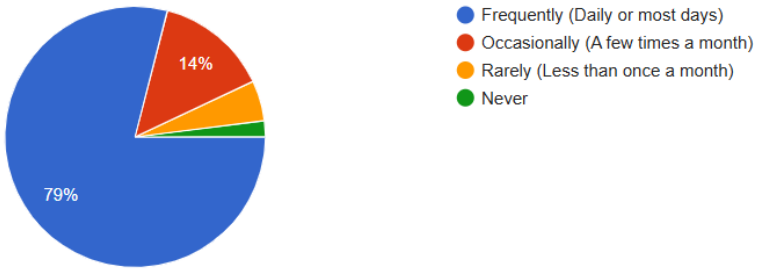


Figure 5.3: Current use of AI tools for exam preparation.

5.2.4 Satisfaction with Current AI Chatbots

The average satisfaction score with current AI chatbots was 3.4 out of 5. Most respondents gave scores of 3 or 4, suggesting that existing AI tools are useful but still imperfect for complete exam preparation. This supports the opportunity for a more specialized and course-specific platform.

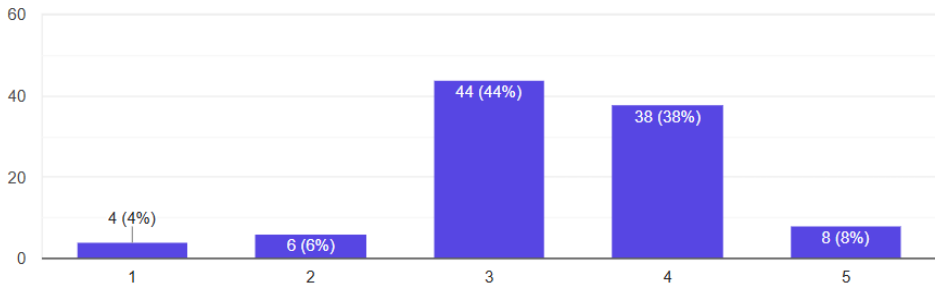


Figure 5.4: Satisfaction with current AI chatbots for studying.

5.2.5 Limitations of Current AI Chatbots

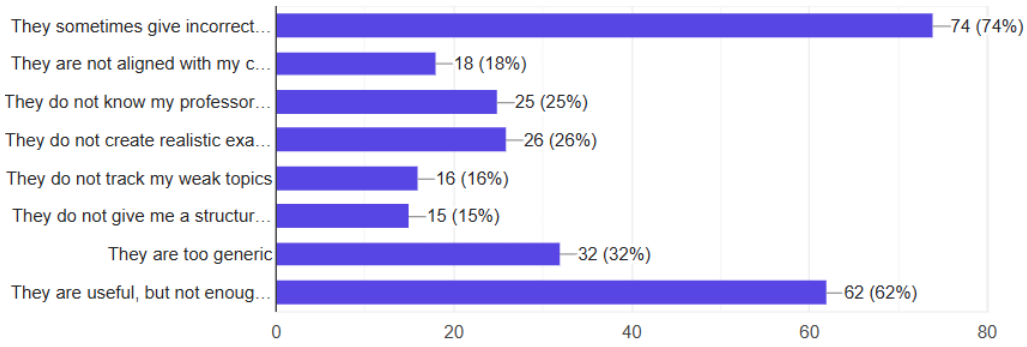


Figure 5.5: Main limitations of current AI chatbots for exam preparation.

Respondents were asked to identify the main limitations of current AI chatbots when used for exam preparation. The survey included the following options:

- They sometimes give incorrect answers;
- They are not aligned with my course materials;

- They do not know my professor's exam style;
- They do not create realistic exams;
- They do not track my weak topics;
- They do not give me a structured study plan;
- They are too generic;
- They are useful, but not enough for complete exam preparation.

The results show that the main limitation of current AI chatbots is accuracy. A total of **74 respondents, stated that AI chatbots sometimes give incorrect answers**, confirming that reliability is a key issue for exam preparation.

The second most selected option was that **AI chatbots are useful, but not enough for complete exam preparation, chosen by 62 respondents**. This suggests that students see AI as helpful, but still incomplete as a structured study tool. Other limitations confirm the need for a more course-specific and structured tool: 32 respondents described AI chatbots as too generic, 26 said they do not create realistic exams, and 25 stated that they do not know the professor's exam style. In addition, 18 respondents reported poor alignment with course materials, while 16 mentioned the lack of weak-topic tracking and 15 the absence of a structured study plan.

5.2.6 Main Difficulty During Exam Preparation

The most common difficulty was **practicing with realistic exam questions, selected by 26% of respondents**. This was followed by a**dapting materials to personal learning needs, selected by 21%**, organizing time, selected by 18%, and understanding whether one is ready for the exam, selected by 17%.



Figure 5.6: Main difficulties faced by students when preparing for university exams.

5.2.7 Feature Validation

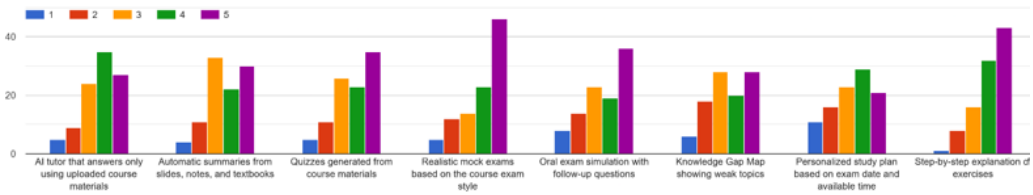


Figure 5.7: Average usefulness score of potential EducationSphere features.

Respondents were asked to rate several potential EducationSphere features from 1 to 5, where 1 means "not useful at all" and 5 means "very useful". The survey included the following features:

- AI tutor based on uploaded course materials;
- Automatic summaries from slides, notes, and textbooks;
- Quizzes generated from course materials;
- Realistic mock exams based on the course exam style;
- Oral exam simulation with follow-up questions;
- Knowledge Gap Map;
- Personalized study plan based on exam date and available time;
- Step-by-step explanation of exercises.

The results show that the most valued feature was **step-by-step explanation of exercises, with an average score of 4.08**. This confirms that students are especially interested in support that helps them understand not only the final answer, but also the reasoning process behind exercises and exam questions.

The second highest-rated feature was **realistic mock exams based on the course exam style, with an average score of 3.93**. This result is consistent with the survey finding that practicing with realistic exam questions is one of the main difficulties faced by students during exam preparation.

Other highly relevant features were **quizzes generated from course materials**, with an average score of 3.72, and the **AI tutor based on uploaded course materials, with an average score of 3.70**. These results suggest that students value interactive practice tools and course-grounded explanations more than generic AI support.

The remaining features received moderately positive scores: automatic summaries scored 3.63, oral exam simulation scored 3.61, the Knowledge Gap Map scored 3.46, and the personalized study plan scored 3.33. Although these features may still be useful, the results suggest that they should remain secondary in the first MVP compared to features directly connected to exercise understanding and realistic exam practice.

5.2.8 Expected Price Compared to Generic AI Chatbots

The survey also explored students' willingness to pay for a course-specific exam-preparation platform. The results show that pricing is a critical issue. While **21% of respondents stated that the platform could cost more than a generic chatbot** if it is clearly better for exams, **31% expected it to cost less than a generic chatbot** and **29% stated that they would not pay for this type of tool**.

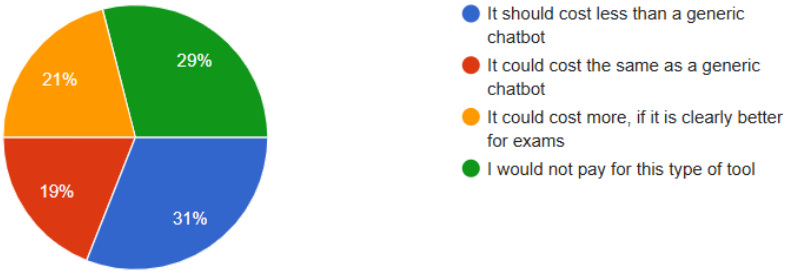


Figure 5.8: Expected cost of a course-specific exam-preparation platform compared to a generic AI chatbot.

5.2.9 Preferred Payment Model

The most preferred payment model was textbf university-provided access, selected by 51% of respondents. The second most preferred option was a **freemium model, selected by 24%**. This indicates that students may be more willing to adopt the platform if the cost is covered by institutions or if a free version is available.



Figure 5.9: Preferred payment model for an exam-preparation app like EducationSphere.

5.3 Validation of the Initial Personas

The survey results confirm that the two initial personas represent relevant student profiles for EducationSphere. Both personas are supported by evidence related to AI usage, dissatisfaction with generic chatbots, exam preparation needs, and interest in course-specific support.

The data strongly validates **Persona 1: The Problem-Solving Learner**. STEM students represented 58% of the sample, and the highest-rated feature was step-by-step explanation of exercises, with an average score of 4.08. This confirms that many students, especially in technical fields, need more than final answers: they need reliable, structured explanations of the reasoning process. Moreover, incorrect answers were the most frequently reported limitation of current AI chatbots, mentioned by 74 respondents. This directly supports the persona's need for trustworthy, course-grounded support.

The survey also validates **Persona 2: The Exam-Ready Student**. The most common difficulty during exam preparation was practicing with realistic exam questions, selected by 26% of respondents. In addition, realistic mock exams based on the course exam style received an average score of 3.93, making them one of the most appreciated

features. This supports the idea that students need tools that reproduce the real exam experience and help them understand whether they are ready.

5.3.1 Limitations of the Survey

Although the survey provides useful evidence, it also has some limitations.

- The sample is strongly represented by STEM students, which may increase the relevance of features such as step-by-step explanations and exercise-based support.
- The survey measures stated preferences rather than actual usage behavior. For this reason, the results should be interpreted as an initial validation signal, not as definitive proof of product-market fit.

Further validation through prototype testing, MVP usage data, and institutional interviews is necessary before making scaling decisions. Based on the validation results, the first MVP should focus on the three strongest signals:

- **step-by-step explanation of exercises**
- **realistic mock exams**
- **quizzes generated from course materials**

Other features, such as personalized study plans and Knowledge Gap Maps, should remain secondary until further behavioral validation confirms stronger user engagement.

5.4 Validation of the Third Persona: The Accessibility-Sensitive Learner

To further validate the third persona, **The Accessibility-Sensitive Learner**, we conducted a semi-structured interview with Lucia Cortino, a second-year Computer Engineering student at the University of Pisa and a blind student. The objective was to understand whether EducationSphere's accessibility-oriented features respond to real barriers experienced by students with specific accessibility needs during university exam preparation. The full interview transcript is reported in Appendix B.1.

The interview showed that, for blind or screen-reader-dependent students, the main problem is not generic simplification of study materials. Instead, the most urgent need is the **accurate transformation of inaccessible academic content into formats that are usable, technically precise, and compatible with assistive technologies.**

- A first relevant insight concerns the difficulty of accessing appropriate study sources. Lucia explained that her main challenge is finding material that is both accessible and clear. She stated that she does not necessarily need a tool that explains everything, but rather one able to identify ***"the synthesis, the core of what I need to prepare for the exam"***. This is particularly important in technical subjects, where formulas, procedures, graphs, and diagrams must be extracted from complex materials without losing their academic meaning.

- The strongest pain point emerged when discussing the effort required to transform slides, PDFs, notes, and books into accessible formats. When asked to rate this effort from 1 to 10, Lucia answered ***"11"***. She also clarified that the issue is not a lack of ability to follow the content, but the presence of an external accessibility barrier: "***it is not that I lose focus because I cannot do it, but because there is an obstacle that goes beyond me***". This distinction is important because it shows that the platform should not treat accessibility as a learning weakness, but as a **technical and structural barrier in the study process**.

- The interview also highlighted which types of materials are most problematic. Lucia identified "images, graphs, formulas" and inaccessible formats, including Latex, as the most difficult materials to use. She explained that formulas written in Latex are not always interpreted correctly by assistive technologies: some symbols may be skipped by text-to-speech tools, for example π or summation signs. In her example, a formula containing a summation, π, and a matrix could be reduced by the screen reader to only "A". This shows that, when mathematical notation is not accessible, the student does not simply receive a less comfortable version of the material; **the meaning of the content may be lost completely**. For this reason, the most strongly validated feature was not a generic summary or a study planner, but a tool capable of explaining formulas, matrices, images, and graphs in an accessible way. When asked to rate this feature, Lucia answered "***10 and a half out of 10"***. This suggests that EducationSphere's Accessible Learning Mode should include a technical accessibility layer able to interpret visual and mathematical content, rather than focusing only on simplified text.

- The interview also confirmed the limits of existing tools. Lucia currently uses text-to-speech tools, AI tools for clarification, and, when possible, support from the university inclusion office. **However, her satisfaction with AI tools remains limited because they may change formatting incorrectly or skip relevant passages**. This supports EducationSphere's positioning as a controlled, course-grounded, and accessibility-aware system rather than a generic chatbot.

- Another important point concerns interface accessibility. Lucia stated that navigating the application with a keyboard and screen reader would be extremely important. She explained that an application becomes frustrating when buttons are not recognized by text-to-speech tools, or when descriptions are not detailed enough or not useful for study purposes. Therefore, EducationSphere should not only generate accessible content but also provide **an interface that is fully compatible with assistive technologies**.

- Finally, the interview provided relevant evidence for the institutional adoption strategy. Lucia stated that she would trust the application more if it were supported

by the university, but especially if it were validated by accessibility experts. She identified ***"accuracy in descriptions"*** and ***"reliability of formulas"*** as the main guarantees needed to use the platform confidently. She also considered a university agreement plausible, especially if EducationSphere helped inclusion offices reduce the manual workload required to adapt books and study materials.

5.4.1 Updates needed for our application

The interview with Lucia Cortino provides strong qualitative validation for the third persona, while also refining its definition. The **Accessibility-Sensitive Learner** should not be interpreted only as a student who needs simplified explanations, micro-goals, or shorter study sessions. For blind or screen-reader-dependent students, the central barrier is the technical inaccessibility of academic materials, especially formulas, graphs, images, diagrams, and non-accessible PDFs.

As a result, EducationSphere's Accessible Learning Mode should prioritize reliable formula interpretation, detailed graph and image descriptions, screen reader compatibility, keyboard navigation, preservation of technical details, and user-controlled accessibility settings. This would make the platform more inclusive and more differentiated from generic AI chatbots. Moreover, the interview supports the B2B/B2B2C strategy, since university endorsement and expert validation could increase trust, while agreements with disability and inclusion offices could represent a realistic path for future pilot projects.

5.5 Additional Validation Methods

In addition to the anonymous survey already conducted, further validation methods could be used to test EducationSphere more deeply. These methods are not implemented at this stage, but they define possible future testing activities to validate the product idea, the main features, and the business model assumptions.

5.5.1 Semi-Structured Interviews with Students

A first possible validation method would be to conduct semi-structured interviews with a smaller group of students. While the survey provided quantitative data, interviews would help us understand the reasons behind students' answers.

The interviews could focus on:

- how students currently prepare for exams;
- which AI tools they already use;
- why they trust or do not trust current AI chatbots;
- which EducationSphere features they would use most frequently;
- what would make them switch from a generic chatbot to a course-specific platform.

This method would be useful to collect qualitative insights and better understand user motivations, frustrations, and expectations.

5.5.2 Clickable Prototype Test

Before building a working MVP, the team could create a clickable prototype using tools such as Figma. The prototype would simulate the main user flow without requiring full technical development.

The test could include tasks such as:

- uploading course materials;
- generating a quiz from uploaded slides;
- asking the AI tutor a question;
- starting a mock exam;
- viewing a Knowledge Gap Map.

The objective would be to evaluate usability, clarity, and feature prioritization. Success would be measured by observing whether users can complete the main tasks easily and whether they understand the platform's value.

5.5.3 MVP Test with a Small Group of Students

A basic MVP could be tested with a small group of students preparing for an actual exam. The MVP should include only the most validated features, such as step-by-step explanations, quizzes from uploaded materials, and realistic mock exams.

The test could measure:

- frequency of use;
- number of uploaded materials;
- number of generated quizzes or mock exams;
- perceived usefulness;
- trust in the answers;
- willingness to continue using the platform.

This would provide stronger evidence than a survey because it would measure real user behavior.

5.5.4 Institutional Validation with Universities and CRUI

Since the survey showed that many students would prefer to use EducationSphere if it were provided or supported by their university, an additional validation stream should focus on institutional adoption. This is important because the platform may not rely only on individual student subscriptions, but could also follow a university-provided access model.

A possible future validation activity would be to contact university representatives, departments, student service offices, innovation offices, or institutional organizations such as "Fondazione CRUI". The objective would not be to implement a partnership immediately, but to understand whether universities would consider EducationSphere valuable, trustworthy, and compatible with their educational objectives.

This validation could focus on:

- what requirements would be necessary for institutional approval;
- what concerns they may have about data privacy, academic integrity, reliability, and fairness;
- whether a pilot project with one course or department would be feasible;
- whether university endorsement would increase student trust and adoption.

5.6 Learning and Decision-Making

The validation process should be treated as a continuous learning cycle. EducationSphere must use survey results, prototype tests, MVP usage, and institutional feedback to decide which features to prioritize, which assumptions to revise, and which business model path to follow.

5.6.1 Success Signals

The main success signals are:

- **Frequent use of core features:** students repeatedly use step-by-step explanations, quizzes from course materials, and realistic mock exams.
- **High perceived usefulness:** users report that EducationSphere helps them understand exercises, practice more effectively, and feel more prepared for exams.
- **Higher trust than generic chatbots:** students perceive answers as more reliable because they are grounded in uploaded course materials.
- **Institutional interest:** at least one university, department, DSA office, or support service shows willingness to test the platform through a pilot.

These signals would support continued investment in the most validated features and confirm that EducationSphere addresses real study needs.

5.6.2 Failure Signals

The main failure signals are:

- **Low usage after first access:** students try the platform once but do not return, suggesting weak perceived value or unclear user experience.
- **Low trust in generated answers:** users still feel the need to verify every answer externally, weakening the value of course-grounded AI.
- **Limited use of mock exams or quizzes:** students do not engage with the features that were most strongly validated by the survey.
- **Weak willingness to pay or weak institutional interest:** students are unwilling to pay individually and universities show no interest in pilots, forcing a revision of the business model.

5.6.3 Evidence-Based Product Decisions

Success and failure signals should guide concrete product decisions.

- If students frequently use step-by-step explanations, quizzes from course materials, and realistic mock exams, these features should become the core of the first MVP.
- If personalized study planning receives lower engagement, it should remain secondary in the first version, even if it may become useful later.

The interview with Lucia also suggests that accessibility should not be treated as a secondary simplification feature, but **as a specific product layer for students who face technical barriers in accessing academic materials**. Therefore, Accessible Learning Mode should prioritize reliable formula interpretation, detailed descriptions of graphs and images, and screen-reader-compatible outputs. These features should be further validated with additional students with accessibility needs and, where possible, with university inclusion offices or accessibility experts.

The same evidence-based logic **applies to the business model.** Since the survey shows a strong preference for university-provided and freemium access, and the interview suggests that university endorsement and expert validation could increase trust among students with accessibility needs, future validation should test whether institutional adoption is more realistic than relying only on individual subscriptions.

Chapter 6

Failure Analysis and Risk Mitigation

6.1 Negative Backcasting Method

The negative backcasting scenario can be formulated as follows:

- EducationSphere launches an MVP and attempts to validate the platform through students and university pilots, but adoption remains weak.
- Students try the platform only once or twice, universities are reluctant to support pilot projects, and the platform is not perceived as sufficiently more valuable than generic AI chatbots.
- As a result, the project fails to convert early interest into repeated usage, institutional contracts, and long-term economic sustainability.

Starting from this scenario, the following sections identify the most relevant causes of failure, the signals that could reveal them early, and the corresponding mitigation strategies.

6.2 Common Failure Causes

6.2.1 Strategic Misalignment

A first potential cause of failure is strategic misalignment. EducationSphere could fail if **the team focuses on too many features at the same time without clearly prioritizing the most validated user needs**. The survey results already suggest that students value some features more strongly than others. If the MVP gives equal priority to less validated features, such as advanced planning tools, complex dashboards, or secondary personalization features, the platform may become difficult to use and may not immediately solve the most urgent student problems.

This risk is particularly relevant because the strongest validation signals point toward practical exam-preparation features, especially **step-by-step explanations, quizzes generated from course materials, and realistic mock exams**. If EducationSphere tries to include too many features in the first version, it may weaken its core value proposition and make the product less clear for early users.

- **Warning Signals**: the platform receives positive survey feedback but weak behavioral engagement, students upload few materials, users register but do not return after the first use, or secondary features are used more for curiosity than for repeated exam preparation.

- **Mitigation**: the first version of the product should remain focused on the **strongest validation signals**. The MVP should prioritize features that create immediate study value, such as generating a quiz from uploaded material, receiving a step-by-step explanation, or practicing with a realistic mock exam. If students do not return after the first use, the team should reduce product complexity and improve the most direct user flows instead of adding new features.

6.2.2 Market Adoption Issues

A second potential cause of failure concerns market adoption. EducationSphere could fail if **students do not perceive a clear difference between a generic AI assistant and a course-specific exam-preparation platform**. Although students already use AI tools for studying, this does not automatically mean that they will adopt a new dedicated platform. Many students may continue using generic chatbots because they are already familiar, free or low-cost, and flexible.

The survey results also show that willingness to pay is uncertain. A significant share of respondents preferred university-provided access or a freemium model. This suggests that **a purely B2C subscription strategy could be risky**, especially in the early stages. If students are interested in the product but unwilling to pay individually, revenue generation may remain weak.

Institutional adoption also presents risks. Universities often have slow decision-making processes and may be cautious about AI tools because of concerns related to academic integrity, reliability, privacy, and compliance. Even if students like the product, EducationSphere may still struggle to obtain official university pilots or annual licences.

- **Warning Signals**: students continue to prefer generic AI chatbots for most study tasks, quizzes and mock exams are generated but rarely completed, or universities express interest in the idea but refuse to support pilot projects.

- **Mitigation**: EducationSphere should follow a **dual adoption strategy**. On the student side, the platform should offer a limited free version to reduce adoption barriers and collect usage data. On the institutional side, the company should use pilot projects to demonstrate measurable value, such as repeated student usage, completed quizzes and mock exams, improved perceived preparedness, and positive feedback from support offices. If individual willingness to pay remains weak, the team should prioritize **freemium access and institutional pilots** rather than relying only on direct student subscriptions.

6.2.3 Technological and Reliability Risks

A third potential cause of failure is technological unreliability. EducationSphere could fail if **AI-generated outputs are inaccurate, misleading, or not sufficiently grounded in the student's uploaded course materials**. This risk is particularly serious in exam preparation, where students need reliable support and may suffer negative academic consequences if they study incorrect material. The survey confirms that incorrect answers are one of the main limitations of current AI chatbots. Therefore, EducationSphere cannot compete only by generating content quickly. It must show that its answers are **course-grounded, transparent, and verifiable**.

Another technological risk concerns the quality of document processing. Students may upload slides, lecture notes, PDFs, or textbooks. If the system cannot correctly extract and interpret these materials, the generated explanations, quizzes, and mock exams may be incomplete or irrelevant. This would reduce the perceived value of the platform and weaken its differentiation from generic AI tools.

This risk is also relevant for accessibility. The interview with Cortino showed that formulas, graphs, images, diagrams, and inaccessible PDFs require accurate interpretation. If the platform simplifies or misreads technical content, it may fail precisely for users who need reliable accessible transformation the most.

- **Warning Signals**: users report low trust in AI-generated explanations, students abandon generated quizzes or mock exams, users return to generic AI chatbots because EducationSphere is not perceived as more reliable, or accessibility-sensitive users report that formulas, graphs, and images are interpreted incorrectly.

- **Mitigation**: the platform should prioritize **course-grounded retrieval, source transparency, and user feedback mechanisms**. EducationSphere should indicate which uploaded materials were used to generate an answer, show references to relevant documents, encourage users to verify critical information, and allow them to flag incorrect or unclear outputs. If users do not trust the answers, the team should improve source grounding, introduce error-reporting mechanisms, and test AI-generated study materials with instructors, tutors, or accessibility experts before scaling.

6.2.4 Ethical Barriers

A fourth potential cause of failure concerns ethical, legal, and regulatory barriers. EducationSphere could fail if **students, professors, or universities perceive the platform as risky from a privacy, copyright, data governance, or academic integrity perspective**. The platform processes student data and uploaded course materials, which creates obligations related to GDPR compliance, intellectual property, and responsible data management.

This is especially relevant because EducationSphere may handle sensitive learning-related data, such as weak topics, quiz performance, study habits, accessibility needs, and exam preparation behavior. If students or universities fear that this information could be misused, adoption may be blocked even if the product is technically useful.

Academic integrity is another important risk. Universities may worry that AI tools encourage students to outsource learning rather than develop understanding. If EducationSphere is perceived as an answer-generation shortcut, it may face resistance from professors and institutions. The platform must therefore be positioned as a **learning support system** rather than as a tool for avoiding study effort.

- **Warning Signals**: privacy, copyright, or academic integrity concerns delay institutional adoption; universities express interest but refuse to move forward with pilot projects; professors perceive the platform as a shortcut for cheating; or students are reluctant to upload course materials and personal learning data.

- **Mitigation**: EducationSphere should implement **privacy-by-design principles, clear consent mechanisms, secure data storage, and transparent policies**

on uploaded materials. The platform should also include academic integrity safeguards, such as explanatory answers, practice-based learning, source grounding, and features that encourage active recall rather than passive copying. If universities hesitate because of compliance concerns, the company should prepare clear documentation on privacy, copyright management, data governance, and academic integrity before entering pilot negotiations.

6.2.5 Business Model and Cost Risks

A further potential cause of failure is an unsustainable business model. EducationSphere could fail if **AI and cloud infrastructure costs grow faster than revenue or validated user engagement**. Since the platform relies on uploaded documents, generated quizzes, mock exams, and AI-powered explanations, usage costs may increase significantly as the number of users grows. If the product is offered at a low price or through a generous free plan without cost controls, the company may generate high operating costs without building a sustainable revenue base.

The institutional model also presents risks. University licences may represent a stronger long-term revenue stream, but **they usually require long sales cycles, compliance checks, and internal approval processes**. If the company depends too early on large institutional contracts, it may run out of resources before converting pilots into paid licences.

- **Warning Signals**: AI and cloud costs increase faster than active usage or revenue, users show strong initial curiosity but limited repeated usage, free users consume expensive features without converting, or institutional pilots do not progress toward paid licences.
- **Mitigation**: EducationSphere should adopt a **staged business model**. In the first phase, the goal should be validation rather than immediate revenue maximization. In the second phase, successful pilots should be converted into small department-level or service-level licences. Only after proving repeated usage and institutional value should the company attempt broader university-wide contracts. Cost control should also be part of the product design: the platform should limit free usage, optimize document processing, monitor AI consumption, and reserve expensive features for premium or institutional users.

6.3 Conclusion

The failure analysis shows that EducationSphere should not focus only on building many AI-based features, but on validating the features that create the clearest value for students. The main risks concern weak adoption, low trust in AI-generated answers, institutional resistance, and unsustainable operating costs.

For this reason, the MVP should prioritize the most validated features: **step-by-step explanations, quizzes generated from course materials, and realistic mock exams**. At the same time, the platform must ensure that its outputs are **course-grounded, transparent, and reliable**, especially for exam preparation and accessibility-related use cases.

EducationSphere should also follow a staged adoption strategy, combining a limited freemium model for students with small institutional pilots. Warning signals such as low

repeated usage, weak trust, poor quiz completion, compliance concerns, or high AI costs should guide future product and business decisions.

In conclusion, a focused and evidence-based approach would reduce the probability of failure and help EducationSphere become a useful, trusted, and sustainable exam-preparation platform.



Figure 6.1: Official EducationSphere's Logo.

Appendix A

Patents

A.1 Patent and Intellectual Property Landscape

The patent and intellectual property landscape was analysed to understand three aspects:

1. The technological areas that may be protected by existing patents,
2. Potential licensing implications for EducationSphere
3. The level of market maturity suggested by patent density.

Since EducationSphere combines several AI-enabled learning functions, the analysis focuses on the main technological components of the platform rather than on a single invention.

A.1.1 Patent Mapping by Platform Component

The most relevant patent areas concern five components of EducationSphere: the Course-Grounded Tutor, Exam Twin AI and Quiz Generation, the Knowledge Gap Map, the Adaptive Study Strategy Engine, and the Accessible Learning Mode. For the **Course-Grounded Tutor**, relevant patents include **WO2025179374A1** and **US11003865B1**, which relate to retrieval-augmented generation and retrieval-augmented language model architectures. This area is important because EducationSphere relies on connecting AI-generated answers to the student's own course materials. For **Exam Twin AI and Quiz Generation**, relevant patents include **US11023684B1**, **CA3055379C**, and **US10467919B2**, related to automatic question generation, AI tutoring, and dynamic assessment. This area is relevant because the platform aims to generate quizzes, mock exams, and assessment activities from course-specific materials. For the **Knowledge Gap Map**, relevant references include **WO2017208083A9**, which concerns learning analytics and diagnostic recommendations and is no longer enforceable. Active patents such as **US10943497B2** and **US11631338B2**, related to deep-learning and transformer-based knowledge tracing. For the **Adaptive Study Strategy Engine**, relevant patents include **US7052277B2**, **US20060166174A1**, and **US11217110B2**. The first two are expired or abandoned and can be used as references for adaptive sequencing and predictive study-path logic. **US11217110B2** is expired for fee-related reasons, but it should still be treated cautiously because active continuations may exist. For the **Accessible Learning Mode**, relevant patents include **US6075968A**, **EP2399251A1**, and **US9672618B1**, related to dyslexia support, speech assistance, and accessible educational interfaces. Since these

patents are expired, ceased, or fee-expired, they can be used as references for simplified explanations, readable formatting, text-to-speech, speech-to-text, and dyslexia-friendly interfaces.

A.1.2 Licensing Strategy and Cost Implications

EducationSphere can use intellectual property strategically to accelerate development, reduce legal uncertainty, and strengthen its competitive position. Based on public royalty-rate benchmarks, licensing costs may vary significantly depending on the strategic importance of the patent, the exclusivity of the agreement, the geographic scope, and the commercial relevance of the technology. The LES 2021 Royalty Survey reports an average royalty rate of approximately 4.82% and a median of 4.75% for high-technology licensing deals [30], while the LES High Tech Sector Royalty Rates and Deal Terms Survey 2017 reports lower software-related royalties, averaging approximately 2.1% [31]. For this reason, a realistic royalty range for AI, software, and EdTech patents can be estimated at approximately **2–7%**. In terms of upfront licensing fees, narrower software or assessment-related patents may require approximately **10k–300k**, while more strategic patents related to AI tutoring, retrieval-augmented architectures, knowledge tracing, or adaptive learning could range from approximately **25k to over 1M**, depending on exclusivity and commercial scope.

A.1.3 Patent Density as a Signal of Market Maturity and Competition

The number of patent query results can be used as an indirect signal of technological maturity and potential competitive intensity. A higher number of patent results suggests that more companies, universities, or inventors have invested in protecting solutions in that area. This may indicate a more mature and crowded market, but also a stronger commercial interest. A lower number of results may suggest a less developed area or a narrower technological niche, although the result should be interpreted cautiously because patent counts depend strongly on the keywords used in the search. The strongest patent density appears in **predictive AI and pedagogical agents**, with approximately **295 results**. This suggests that AI-based tutoring and intelligent pedagogical support are relatively mature and competitive areas. **Adaptive sequencing** shows approximately **60 results**, while **automated question generation** shows approximately **55 results**, indicating significant interest in personalized learning paths and AI-generated assessment. These areas are strategically important for EducationSphere but may also involve higher competitive and IP risk. Medium-density areas include **accessibility and OCR support for learning disabilities**, with approximately **32 results**, **AI-supported lesson planning**, with approximately **24 results**, and **learning analytics**, with approximately **15 results**. These results suggest active but less saturated spaces, where EducationSphere may find opportunities to differentiate through user experience, accessibility, and study-progress insights. Lower-density areas include **retrieval-augmented language models**, with approximately **9 results**, **knowledge tracing**, with approximately **3–14 results**, **dyslexia-oriented speech recognition**, with approximately **2 results**, **dyslexia screening**, with approximately **4 results**, and **candidate knowledge item extraction**, with approximately **1 result**. These lower counts should not be interpreted as weak market demand. Instead, they may reflect newer technologies, narrower patent

terminology, or search-query limitations. For EducationSphere, these areas may represent opportunities for differentiation, especially if the platform combines them into a coherent study workflow rather than treating them as isolated features.

A.2 Queries

Patent	**Query**
WO2025179374A1	"synthetic question embeddings" "retrieval-augmented generation"
US11003865B1	"retrieval-augmented language model" "pre-training" "fine-tuning"
US11023684B1	"automatic generation of questions from text"
CA3055379C	"automated question generation"
US10467919B2	"lesson plan" "AI engine"
WO2017208083A9	"learning analytics" ("on-task" OR "off-task") students
US10943497B2	"personalized e-learning" "knowledge tracing"
US11631338B2	"knowledge tracing" "learner interactions"
US7052277B2	(stimulus OR stimuli) AND ("measuring accuracy" OR "response time") AND ("sequencing algorithm" OR "dynamically adapts")
US20060166174A1	("predictive artificial intelligence" OR "pedagogical agent")
US11217110B2	"digital source material" "candidate knowledge items"
US6075968A	("learning-disabled" OR "learning disabled" OR "dyslexia" OR "dyslexic") AND ("character-recognized" OR "character recognition" OR "OCR") AND ("work station" OR "workstation" OR "setup")
EP2399251A1	"speech recognition system" "assist persons having dyslexia"
US9672618B1	"dyslexia screening" ("gaze movement" OR "eye tracking" OR "pupil") (multimedia OR multimodal)

Table A.1: Patent landscape queries for EducationSphere-related technologies

Appendix B

Full Interview

B.1 Validation of the Third Persona: Full Interview

Question 1: During the preparation of a university exam, what are the main difficulties you encounter while studying? Lucia explained that the main issue is finding study sources that are both accessible and clear. She stated that she does not necessarily need an application that explains everything, but rather a tool able to identify "the synthesis, the core of what I need to prepare for the exam". This is especially relevant in technical subjects, where formulas, procedures, and essential steps must be extracted from complex materials without forcing the student to navigate an entire book or inaccessible document.

This answer validates a central assumption of EducationSphere: students do not only need more content, but better structured and more usable content. For Lucia, the platform would create value if it helped extract the relevant parts of the course material while preserving the technical meaning needed for exam preparation.

Question 2: How much does it weigh on you to transform slides, PDFs, notes, or books into a format that is more suitable for your way of studying, from 1 to 10? Lucia answered "11". This was one of the clearest signals of the interview. The burden is not only the study itself, but the preliminary work required to make the study material accessible.

When asked whether she loses focus while studying, Lucia clarified that the problem is not her inability to follow the content. She stated that "it is not that I lose focus because I cannot do it, but because there is an obstacle that goes beyond me". This distinction is important: the barrier is not cognitive weakness, but the inaccessibility or poor adaptation of the material.

Question 3: Which types of university materials are most difficult for you to use? Lucia answered that the most problematic materials are "images, graphs, formulas" and "some formats that are not accessible", including Latex.

This is one of the strongest validation points for the third persona. EducationSphere's Accessible Learning Mode should not focus only on readable text or shorter summaries. It should also support technical content that is often inaccessible to blind students, such as mathematical formulas, graphs, diagrams, matrices, and visual explanations.

Question 4: What problems do formulas and Latex create with current assistive technologies? We asked whether the difficulty depends on the fact that images and formulas are not properly explained. Lucia gave a concrete example: "if you write formulas in Latex, the symbols are not always transcribed" and, even when they are present, "the text-to-speech cannot read them". She explained that the screen reader may skip essential symbols: "it skips pieces of the formulas, for example, it does not read π, it does not read the summation". In her example, a formula containing a summation, π, and a matrix could be reduced by the screen reader to only "A".

This answer is highly relevant for EducationSphere because it identifies a precise technical problem that generic AI chatbots and standard text-to-speech tools do not solve reliably. If formulas are read incorrectly, the student does not simply receive a less comfortable version of the material; the academic meaning of the content may be lost completely.

Question 5: Would an application that can explain mathematical formulas, matrices, images, and graphs in an accessible way be useful for you? Rate the feature from 1 to 10. Lucia answered "10 and a half". This was the strongest feature-validation signal in the interview. It suggests that an accessibility-oriented function capable of interpreting formulas, graphs, images, and technical diagrams would be significantly more valuable than generic summaries or generic study planning.

For EducationSphere, this means that the Accessible Learning Mode should include a technical accessibility layer, not only a simplified-reading mode.

Question 6: In what way does your accessibility need influence your anxiety before exams? Lucia explained that anxiety increases mainly when an exam is not accessible. She mentioned an Operations Research exam involving integer linear programming, graphs, and trees, explaining that the accessibility barrier can influence performance because the difficulty is "not due to my knowledge", but to the obstacle created by the inaccessible format.

This insight is important because it shows that accessibility affects not only comfort, but also perceived fairness and exam performance. If a student understands the subject but cannot access graphs, formulas, or diagrams properly, the exam result may reflect an accessibility barrier rather than actual preparation.

Question 7: Which tools do you currently use to study better or make materials more accessible? Lucia mentioned text-to-speech tools, AI tools used for clarification, and, when possible, the university inclusion office. She rated her average satisfaction with text-to-speech tools as 7 out of 10, but her satisfaction with AI tools as only 5 out of 10.

She explained that AI tools can be useful for clarifications, but are less reliable when asked to change formatting or adapt technical materials. In her words, AI may "modify the formatting further" and consequently "skip some passages". This supports EducationSphere's positioning as a controlled, course-grounded and accessibility-aware system rather than a generic chatbot.

Question 8: Are PDFs, slides, and materials provided by professors easily readable with screen readers? Would it be useful if the application transformed visual materials into clear textual descriptions? Lucia answered that,

on average, she would rate the accessibility of university materials as 5 out of 10. This indicates that PDFs, slides, and other materials provided by professors are only partially accessible and often require additional adaptation before they can be used effectively.

When asked whether it would be useful if the application transformed visual materials into clear textual descriptions, Lucia answered positively. This directly supports a possible EducationSphere feature focused on transforming images, graphs, and diagrams into accessible textual explanations.

Question 9: How important would it be for you to navigate the application only with keyboard or screen reader? Lucia answered: "Very much". She also explained that an application becomes frustrating when some functions are not accessible with screen readers, for example when "the button is not recognized by the text-to-speech", or when descriptions are "not detailed enough" or "not useful for study purposes".

This point is crucial for product development. EducationSphere cannot claim to support accessibility only by generating accessible content. The interface itself must be accessible, navigable by keyboard, and compatible with screen readers. Otherwise, the application would reproduce the same barriers it aims to solve.

Question 10: Would you prefer the application to explicitly ask which adaptations you need, instead of automatically inferring them? Lucia answered: "Better to ask explicitly". She also rated the possibility of manually modifying how the app adapts content as 8.5 out of 10.

This suggests that EducationSphere should not impose a fixed accessibility profile. Instead, it should allow users to configure their needs, choose output formats, and adjust the level of detail. This is especially important because accessibility needs vary significantly between students.

Question 11: Would you trust the application more if it were provided by the university or validated by accessibility experts? Could a university agreement be realistic? We asked whether Lucia would trust the application more if it were provided or approved by the university. She answered yes. When asked whether she would trust it more if it were validated by a disability office or by accessibility experts, she answered: "Accessibility experts". When asked what guarantees would be necessary to use the application confidently, she identified "accuracy in descriptions" and "reliability of formulas".

This answer is aligned with EducationSphere's B2B/B2B2C logic. University endorsement can increase trust, but accessibility validation should involve real experts and real users. For the platform to be credible, it must demonstrate that its outputs are not only convenient, but accurate and technically reliable.

The interview also explored whether a university agreement could be realistic. Lucia considered this plausible, especially because the university already provides accessibility-related services, such as material transcription and specific agreements for students with disabilities. She explained that the inclusion office already works on adapting books and study materials and may be trying to automate part of this process. Therefore, EducationSphere could be interesting if it reduced manual workload: the university might prefer to partially pay for a service that allows staff to focus on other activities.

This provides a relevant institutional insight. EducationSphere could create value not only for individual students, but also for university disability and inclusion offices by

reducing the time and effort required to make study materials accessible.

Question 12: Which concrete problem should this application solve for you to say that it is truly useful? Lucia answered that the most important function would be the description of images and graphs "without losing technical details". She explained that, in engineering, it is not enough to say that "there is a graph with a line". Technical descriptions must preserve the information required to understand the subject.

She added that, in mathematical and scientific subjects, it is necessary "not to lose the details", and that the application should also make formulas accessible and provide effective summaries when needed. When asked what she would change in her current study experience, she answered: "accessibility in general, not only of the material, but also of development environments".

Bibliography

[1] P. Vaishnavi, A. Manaswi, G. Kirti, D. Vaishnavi, and Y. N. Jadhav, "Smart study planner-ai based personalized time management," *International Journal of Engineering Development and Research (IJEDR)*, vol. 14, no. 1, pp. 116–120, 2026. [Online]. Available: https://rjwave.org/ijedr/papers/IJEDR2601268.pdf

[2] R. Sajja, Y. Sermet, B. Fodale, and I. Demir, "Evaluating ai-powered learning assistants in engineering higher education with implications for student engagement, ethics, and policy," *Scientific Reports*, 2026. [Online]. Available: https://pmc.ncbi.nlm.nih.gov/articles/PMC12932848/

[3] J. Wu, Y. Wang, F. Chen, X. Yin, and Y. He, "Impact of ai-powered adaptive learning platforms on english reading proficiency: Evidence from structural equation modeling," *IEEE Access*, vol. 13, pp. 88 230–88 242, 2025. DOI: 10.1109/ACCESS.2025.3571055

[4] Luke, *The impact of generative ai and rag on personalized learning*, *Makebot.ai*, Accessed: 2026-05-08, Oct. 2024. [Online]. Available: https://www.makebot.ai/blog-en/the-impact-of-generative-ai-and-rag-on-personalized-learning

[5] I. S. Nor Afian, L. Salahuddin, A. Idris, and F. Abdul Rahim, "Ai-powered tutoring system for automatic notes summarization and quiz generation," *International Journal of Research and Innovation in Social Science*, vol. 9, no. 10, 2025. DOI: 10.47772/IJRISS.2025.910000097

[6] C. Sadler, *The future of ai tutoring in higher ed*, *New America*, Accessed: 2026-05-08, Apr. 2023. [Online]. Available: https://www.newamerica.org/insights/the--future-of-ai-tutoring-in-higher-ed/

[7] F. Guo, L. Zhang, T. Shi, and H. Coates, "Whether and when could generative ai improve college student learning engagement?" *Behavioral Sciences*, vol. 15, no. 8, 2025, ISSN: 2076-328X. DOI: 10.3390/bs15081011 [Online]. Available: https://www.mdpi.com/2076-328X/15/8/1011

[8] O. V. Sergeeva, M. R. Zheltukhina, T. Shoustikova, L. R. Tukhvatullina, D. A. Dobrokhotov, and S. V. Kondrashev, "Understanding higher education students' adoption of generative ai technologies: An empirical investigation using utaut2," *Contemporary Educational Technology*, vol. 17, no. 2, ep571, 2025. DOI: 10.30935/cedtech/16039 [Online]. Available: https://www.cedtech.net/article/understanding-higher-education-students-adoption-of-generative-ai-technologies-an-empirical-16039

[9] B. Ward, D. Bhati, F. Neha, and A. Guercio, *Analyzing the impact of ai tools on student study habits and academic performance*, 2024. arXiv: 2412.02166 [cs.AI]. [Online]. Available: https://arxiv.org/abs/2412.02166

[10] Dimension Market Research, *Artificial intelligence (ai) in edtech market*, 2025. [Online]. Available: https://dimensionmarketresearch.com/report/ai-in-edtech-market/

[11] EdTech Insiders, *Ai startup market maps: Edtech insiders*, Feb. 2026. [Online]. Available: https://publicservicesalliance.org/2026/02/18/ai-startup-market-maps-edtech-insiders/

[12] Porter's Five Forces, *What is competitive landscape of chegg company?* Nov. 2025. [Online]. Available: https://portersfiveforce.com/blogs/competitors/chegg

[13] Moonlight, *Analyzing the impact of ai tools on student study habits and academic performance*, Mar. 2025. [Online]. Available: https://www.themoonlight.io/en/review/analyzing-the-impact-of-ai-tools-on-student-study-habits-and-academic-performance

[14] Knowunity, *Ai at knowunity*, Accessed: 2026-05-30, 2026. [Online]. Available: https://knowunity.com/about/knowunity-ai

[15] A. Shuhaiber, "Chatgpt in higher education: A student's perspective," *Journal of Open Innovation: Technology, Market, and Complexity*, 2025.

[16] Estha, *Ai tutors vs human tutors: Complete cost and effectiveness comparison*, Accessed: 2026-05-30, 2026. [Online]. Available: https://estha.ai/blog/ai-tutors-vs-human-tutors-complete-cost-and-effectiveness-comparison/

[17] M. P. Singh, *How to build an e-learning platform*, https://engineerbabu.com/blog/how-to-build-an-e-learning-platform/, Accessed: 2026-05-26, EngineerBabu, Feb. 2026.

[18] CloudZero, *LLM api pricing comparison in 2026: Every major model, ranked by cost*, https://www.cloudzero.com/blog/llm-api-pricing/, Accessed: 2026-05-26, CloudZero, May 2026.

[19] Team DigitalA11Y, *Web accessibility audit cost guide: Detailed pricing factors 2026*, https://www.digitala11y.com/how-much-does-a-web-accessibility-audit-cost/, Accessed: 2026-05-26, DigitalA11Y, Feb. 2026.

[20] ContractsCounsel, *SaaS agreement pricing: Average costs and lawyer bids*, https://www.contractscounsel.com/b/saas-agreement-cost, Accessed: 2026-05-26, ContractsCounsel, May 2026.

[21] WordStream. "How much do instagram ads cost in 2026?" Accessed: 2026-06-03. [Online]. Available: https://www.wordstream.com/blog/instagram-ads-cost

[22] AdManage. "Instagram ads cost in 2026: Cpc, cpm + roi benchmarks." Accessed: 2026-06-03. [Online]. Available: https://admanage.ai/blog/how-much-do-instagram-ads-cost

[23] AdManage. "Tiktok ads cost: Minimum budget, cpc & cpm rates." Accessed: 2026-06-03. [Online]. Available: https://admanage.ai/blog/tiktok-ads-cost

[24] Shopify. "Influencer pricing: The cost of influencers in 2026." Accessed: 2026-06-03. [Online]. Available: https://www.shopify.com/blog/influencer-pricing

[25] GrowSurf, *Best mobile app referral program examples*, Accessed: 2026-06-07, 2026. [Online]. Available: https://growsurf.com/examples/mobile-app-referral-programs/

[26] Eurostat, *Tertiary education statistics*, https://ec.europa.eu/eurostat/statistics-explained/index.php?title=Tertiary_education_statistics, Accessed: 2026-06-02, 2025.

[27] European Commission, *Higher education initiatives*, https://education.ec.europa.eu/education-levels/higher-education/about-higher-education, Accessed: 2026-06-02, 2025.

[28] Ministero dell'Università e della Ricerca, *Ustat: I numeri della formazione*, https://ustat.mur.gov.it/, Accessed: 2026-06-02, 2025.

[29] ANVUR, *Rapporto 2026 sul sistema della formazione superiore e della ricerca*, https://www.anvur.it/it/novita-ed-eventi/anvur-pubblica-il-rapporto-2026-sul-sistema-della-formazione-superiore-e-della, Accessed: 2026-06-02, 2026.

[30] IPWatchdog, *Les 2021 royalty survey reports: Licensing market update and a look back at the les royalty valuation method*, https://ipwatchdog.com/2022/10/03/les-2021-royalty-survey-reports-licensing-market-update-look-back-les-royalty-valuation-method-making/, Accessed: 2026-05-26, 2022.

[31] Licensing Executives Society USA and Canada, *Les high tech sector royalty rates and deal terms survey 2017*, https://cdn.ymaws.com/members.lesusacanada.org/resource/collection/4C61AEDD-DEA7-4BAD-AA71-82F7FB8005EF/HTS_SurveyPPT_2017_-_Mar29_FINAL.pdf, Accessed: 2026-05-26, 2018.